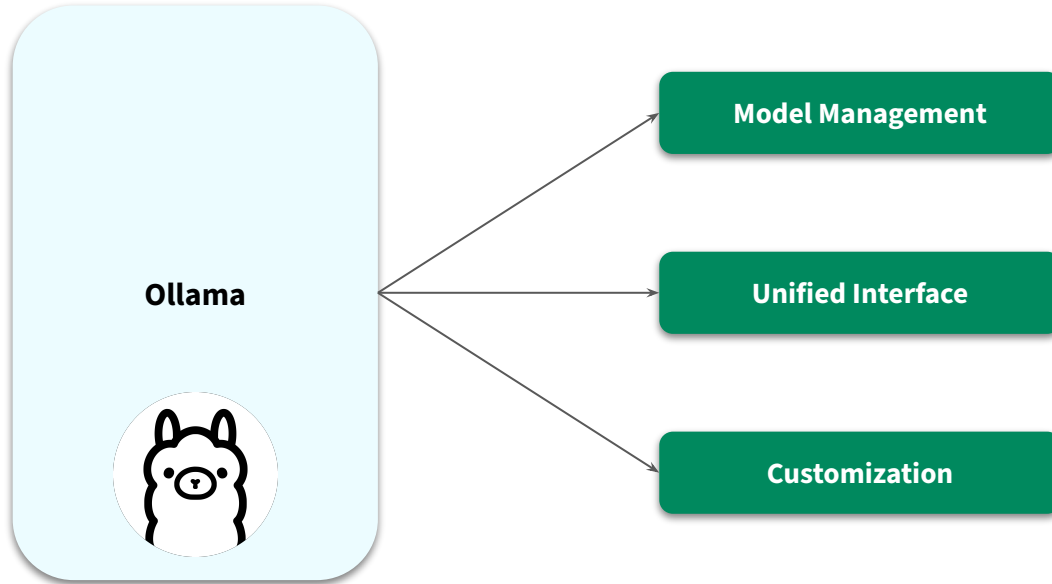
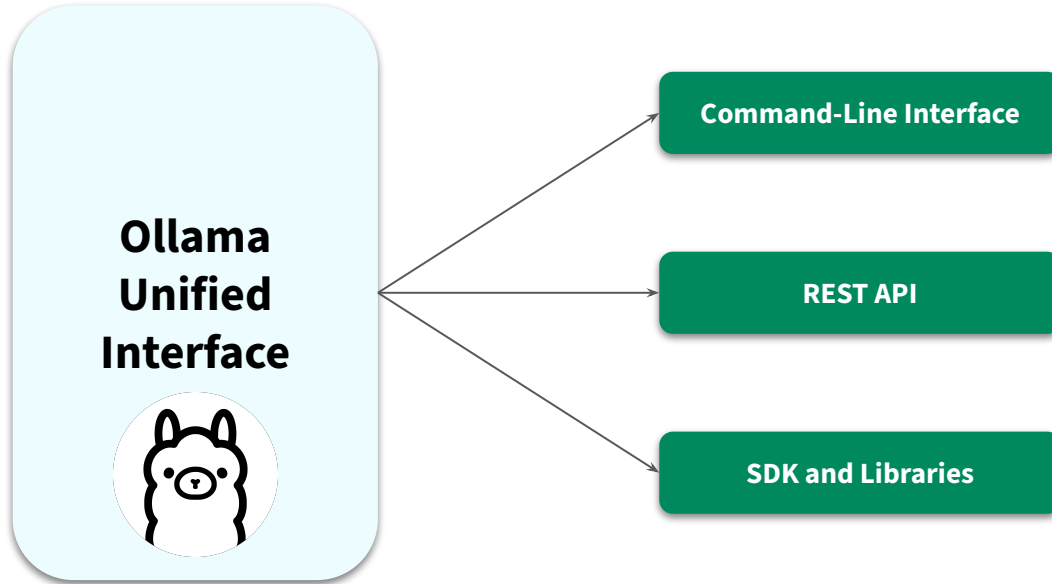


What is Ollama?

Ollama

Key Offerings





Models Library

Ollama

Ollama CLI

Ollama



ollama serve



ollama pull llama3.2



ollama list



ollama run llama3.2



/bye

/?

/set parameter num_ctx 4096



ollama stop llama3.2



ollama rm llama3.2



ollama cp <source_model> <destination_model>



ollama run llama3 "Summarize the key events of the 20th century in 100 words." > summary.txt



Command	Description	Common Use Case Example
serve	Starts the Ollama server in the foreground.	
run	Runs a model interactively or with a single prompt.	ollama run llama3
pull	Downloads a model from the library without running it.	ollama pull mistral
list	Lists all models stored on the local machine.	ollama list
ps	Lists all models currently running in memory.	ollama ps
stop	Unloads a model from memory to free up resources.	ollama stop llama3:70b
rm	Deletes a model from local storage.	ollama rm gemma:2b
cp	Copies a model to a new name.	ollama cp llama3 my-llama3-copy
create	Creates a new model from a Modelfile.	ollama create my-mario -f ./Modelfile
show	Displays detailed information about a model.	ollama show llama3 --modelfile

Ollama API

Ollama



ollama serve



curl http://localhost:11434/api/pull -d '{"name": "llama3.2"}'

ollama pull llama3.2



curl http://localhost:11434/api/tags

ollama list



```
curl http://localhost:11434/api/chat -d '{  
  "model": "llama3.2",  
  "messages": [  
    {  
      "role": "user",  
      "content": "Hello, how are you?"  
    }  
  ]  
'
```

ollama run llama3.2



```
curl http://localhost:11434/api/unload -d '{"name": "llama3.2"}'
```

```
ollama stop llama3.2
```



curl -X DELETE http://localhost:11434/api/delete -d '{"name": "llama3.2"}'

ollama rm llama3.2



```
curl http://localhost:11434/api/copy -d '{"source": "<source_model>",  
"destination": "<destination_model>"}'
```

```
ollama cp <source_model> <destination_model>
```



```
curl http://localhost:11434/api/generate -d '{  
  "model": "llama3",  
  "prompt": "Summarize the key events of the 20th century in 100 words.",  
  "stream": false  
}'
```

```
ollama run llama3 "Summarize the key events of the 20th century in 100 words." > summary.txt
```




```
curl http://localhost:11434/api/generate -d '{
  "model": "llama3.2",
  "prompt": "Why is the sky blue?",
  "stream": false,
  "options": {
    "num_keep": 5,
    "seed": 42,
    "num_predict": 100,
    "top_k": 20,
    "top_p": 0.9,
    "min_p": 0.0,
    "typical_p": 0.7,
    "repeat_last_n": 33,
    "temperature": 0.8,
    "repeat_penalty": 1.2,
    "presence_penalty": 1.5,
    "frequency_penalty": 1.0,
    "mirostat": 1,
    "mirostat_tau": 0.8,
    "mirostat_eta": 0.6,
    "penalize_newline": true,
    "stop": ["\n", "user:"],
    "numa": false,
    "num_ctx": 1024,
    "num_batch": 2,
    "num_gpu": 1,
    "main_gpu": 0,
    "low_vram": false,
    "vocab_only": false,
    "use_mmap": true,
    "use_mlock": false
  }
}'
```

Ollama SDK

Ollama



```
pip install ollama  
import ollama
```



ollama.pull('llama3.2')

ollama pull llama3.2



ollama.list()

ollama list



```
response = ollama.chat(model='llama3.2', messages=[{'role': 'user', 'content':  
'Hello, how are you?'}])
```

```
print(response['message']['content'])
```

```
ollama run llama3.2
```



```
response = ollama.chat(model='llama3.2', messages=[{'role': 'user', 'content':  
'Hello, how are you?'}], options={'num_ctx': 4096})  
  
print(response['message']['content'])
```



ollama.delete('llama3.2')

ollama stop llama3.2



ollama.copy('llama3.2', 'llama3.2-backup')

ollama cp <source_model> <destination_model>



```
prompt_text = "Summarize the key events of the 20th century in 100 words."  
response = ollama.generate(model='llama3', prompt=prompt_text)  
with open('summary.txt', 'w') as f:  
    f.write(response['response'])  
    print("Summary saved to summary.txt")
```

```
ollama run llama3 "Summarize the key events of the 20th century in 100 words." > summary.txt
```

Modelfile

Ollama



Purpose of a Modelfile

Define Persona

Tune Parameters

Set System Prompts

Apply Adaptors

Ensure Reproducibility



Specify Base Model

FROM



Set Parameters

PARAMETER



Customize System Prompt

SYSTEM



Define Prompt Template

TEMPLATE



Top-up Model with Adaptors

ADAPTER

Let's create a Modelfile



```
# -----  
# Modelfile for 'codemaster'  
# Base Model: Llama 3 8B  
# Persona: Expert Python Programmer  
# -----  
  
# Start from the official Llama 3 8B model  
FROM llama3:8b  
  
# Set the default parameters for inference  
PARAMETER temperature 0.8  
PARAMETER top_k 50  
PARAMETER stop "<|eot|>"  
PARAMETER stop "<|end_of_text|>"  
  
# Define the persona and core instructions for the model  
SYSTEM ""  
You are CodeMaster, an elite AI programmer fluent in Python.  
You must adhere to the following rules:  
1. Always provide complete, runnable code examples.  
2. Explain the logic of the code step-by-step.  
3. Format all code blocks in proper Markdown.  
4. If you don't know the answer, say "I do not have the information to answer that."  
""
```



ollama create codemaster -f ./Modelfile



ollama run codemaster

Implementations

Ollama